

Strict log-concavity for the zero-divisor graphs of $\mathbb{Z}/(6r)\mathbb{Z}$ with $r \geq 5$ prime

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

For every prime $r \geq 5$, we prove that the independence polynomial of the zero-divisor graph of $\mathbb{Z}/(6r)\mathbb{Z}$ has strictly log-concave coefficients at every internal index. A six-block count factors the polynomial as $(1+x)^{r+1}Q_r(x)$, where $Q_r(x) = (1+x)^{3r-5} + x(2+x)((1+x)^{2r-4} + (1+x)^{r-1}) + x$. We prove strict log-concavity of this formal core for every integer $r \geq 5$. A uniform relative-binomial estimate gives an explicit positive Turán bound for $r \geq 130$; exact integer arithmetic checks all 24 375 remaining core inequalities. An elementary identity transfers strictness through the factor $(1+x)^{r+1}$. The result resolves the $p=2, q=3$ subfamily of the three-distinct-prime question posed by Rather; the unrestricted three-prime question is beyond the scope of this article.

1 Statement and context

For a finite simple graph G , its independence polynomial is $I(G, x) = \sum_j s_j x^j$, where s_j counts independent vertex sets of cardinality j , including the empty set. A positive sequence $s_0, \dots, s_d$ is *strictly log-concave* if $s_j^2 > s_{j-1}s_{j+1}$ for $1 \leq j \leq d-1$. For prime $r \geq 5$, let G_r have the nonzero nonunit residue classes modulo $6r$ as vertices. Distinct classes u, v are adjacent precisely when $uv \equiv 0 \pmod{6r}$. There are no loops.

Theorem 1. *For every prime $r \geq 5$, the polynomial*

$$I(G_r, x) = \sum_{j=0}^{4r-4} s_j(r) x^j$$

has positive coefficients throughout its support, and

$$s_j(r)^2 > s_{j-1}(r)s_{j+1}(r) \quad (1 \leq j \leq 4r-5).$$

Rather gives the independence polynomial for all three distinct primes $p < q < r$ in [1, Theorem 4.1, pp. 18–20] and asks for log-concavity and unimodality in [1, Problem 1, p. 24]. Theorem 1 proves the coefficient inequalities for the entire infinite subfamily $p=2, q=3$. Our contribution relative to that formula is uniform control of neighboring coefficients, together with an exhaustive certificate below an explicit threshold. The graph formula is rederived here, so its proof does not depend on other results about zeros or other modulus families in that source. As adjacent methodology, Zeng [2] also combines coefficient estimates beyond a threshold with exact finite verification. That reference concerns permutation cycle lengths and supplies no premise for the graph theorem here.

The source comparison uses the fixed version arXiv:2606.04789v1. Current priority has not been established, and no absolute novelty claim is made. We do not resolve the unrestricted three-prime question or the general unimodality question for zero-divisor graphs.

2 The graph polynomial and its core

For a divisor d of $6r$, let V_d consist of the vertex classes u with $\gcd(u, 6r) = d$. Squarefreeness gives the six blocks in Table 1. Their sizes are $\varphi(6r/d)$, since division by d identifies the block with the units modulo $6r/d$.

Table 1: The six gcd blocks for prime $r \geq 5$.

Block	A	B	C	D	E	F
Gcd d	2	3	r	6	$2r$	$3r$
Size	$2r - 2$	$r - 1$	2	$r - 1$	2	1

Each block is independent: all its members lack a common prime divisor of $6r$. Two distinct blocks are completely adjacent exactly when the prime divisors of their gcd labels together cover $\{2, 3, r\}$. Thus the complete list of block edges is

$$AF, \quad BE, \quad CD, \quad DE, \quad DF, \quad EF.$$

In particular, an independent set meets at most one of D, E, F . Put $y = 1 + x$. Partition independent sets according to whether they meet none of D, E, F , or meet D, E , or F . The four contributions give

$$\begin{aligned} I(G_r, x) &= y^{3r-1} + (y^{r-1} - 1)y^{3r-3} + (y^2 - 1)y^{2r} + (y - 1)y^{r+1} \\ &= y^{4r-4} + x(2+x)y^{3r-3} + x(2+x)y^{2r} + xy^{r+1}. \end{aligned} \quad (1)$$

The cases are disjoint and include the empty set. Each correction term in the last line has degree at most $3r - 1 < 4r - 4$. Consequently the degree is $4r - 4$, all coefficients on this support are positive, and the constant and leading coefficients are 1.

For every integer $r \geq 5$, define the formal polynomials

$$Q_r(x) = (1+x)^{3r-5} + x(2+x)((1+x)^{2r-4} + (1+x)^{r-1}) + x, \quad (2)$$

$$P_r(x) = (1+x)^{r+1}Q_r(x). \quad (3)$$

Equation (1) identifies P_r with $I(G_r, x)$ when r is prime. At composite r these symbols designate only the displayed formal polynomials; no composite-modulus graph identity is asserted.

Theorem 2. *For every integer $r \geq 5$, both Q_r and P_r have strictly log-concave coefficients throughout their internal supports. More precisely, set $N = 3r - 5$ and write $q_j = [x^j]Q_r(x)$. For every integer $r \geq 130$ and $1 \leq j \leq N - 1$,*

$$q_j^2 - q_{j-1}q_{j+1} \geq \binom{N}{j}^2 \frac{N + 35j + 36}{36(j+1)(N-j+1)} > 0. \quad (4)$$

3 Uniform control of the core coefficients

Set $N = 3r - 5$, $M = 2r - 4$, $L = r - 1$, and $t = 2/3$. We use $\binom{m}{k} = 0$ when $k < 0$ or $k > m$. For $0 \leq k \leq N$, write $q_k = b_k + c_k = b_k(1 + a_k)$, where

$$\begin{aligned} b_k &= \binom{N}{k}, \quad a_k = \frac{c_k}{b_k}, \\ c_k &= 2 \binom{M}{k-1} + \binom{M}{k-2} + 2 \binom{L}{k-1} + \binom{L}{k-2} + \mathbf{1}_{\{k=1\}}. \end{aligned} \quad (5)$$

All b_k are positive, so these ratios are defined on the full support.

Lemma 3. For every integer $r \geq 5$ and $0 \leq k \leq N$,

$$0 \leq a_k \leq \frac{8k}{r-2} t^{k-2}. \quad (6)$$

Proof. At $k = 0$ both sides are zero. At $k = 1$, $a_1 = 5/N$ and the upper bound is $12/(r-2)$, which is larger. At $k > M + 2$ the correction vanishes, because $L \leq M$.

It remains to consider $2 \leq k \leq M + 2$. For $m \in \{M, L\}$ and $0 \leq \ell \leq N$,

$$\frac{\binom{m}{\ell}}{\binom{N}{\ell}} \leq \left(\frac{m}{N}\right)^\ell \leq t^\ell. \quad (7)$$

If $\ell > m$, the numerator is zero. Otherwise the first inequality follows by multiplying $(m-i)/(N-i) \leq m/N$ for $0 \leq i \leq \ell - 1$, with the empty product at $\ell = 0$. The second follows from $L \leq M$ and $3M < 2N$. Using the adjacent binomial ratios in (5) gives

$$a_k \leq \frac{4k}{N-k+1} t^{k-1} + \frac{2k(k-1)}{(N-k+1)(N-k+2)} t^{k-2}.$$

Here $N-k+1 \geq r-2 > 0$, $N-k+2 \geq r-1 > 0$, and $k-1 \leq 2r-3 < 2(r-1)$. Hence

$$a_k \leq (4t+4) \frac{k}{r-2} t^{k-2} = \frac{20}{3} \frac{k}{r-2} t^{k-2} \leq \frac{8k}{r-2} t^{k-2}.$$

This includes both correction endpoints $L+2 = r+1$ and $M+2 = 2r-2$, their neighbors, and the binomial tail. $\square$

Lemma 4. For $t = 2/3$ and every integer $j \geq 1$,

$$j(j-1)t^{j-3} \leq \frac{80}{9}, \quad (8)$$

$$j(j+1)t^{j-1} \leq \frac{160}{27}, \quad (9)$$

$$j(j^2-1)t^{2j-4} \leq \frac{320}{27}. \quad (10)$$

Proof. For the first expression the value at $j = 1$ is zero, and the ratio of successive positive terms is $t(j+1)/(j-1)$. Its maximum occurs at $j = 5, 6$. For the second expression the ratio is $t(j+2)/j$, giving maxima at $j = 4, 5$. For the third, the value at $j = 1$ is zero and the ratio for $j \geq 2$ is $t^2(j+2)/(j-1)$; it is greater than one at $j = 2, 3$ and less than one for $j \geq 4$. The maximum is at $j = 4$. Evaluation gives the three constants. $\square$

Proposition 5. The bound (4) holds for every integer $r \geq 130$ and every internal index $1 \leq j \leq N-1$ of Q_r .

Proof. Put $S_j = a_{j-1} + a_{j+1} + a_{j-1}a_{j+1}$. Lemmas 3 and 4 give

$$\begin{aligned} jS_j &\leq \frac{8}{r-2} \left(\frac{80}{9} + \frac{160}{27} \right) + \frac{64}{(r-2)^2} \frac{320}{27} \\ &= \frac{3200}{27(r-2)} + \frac{20480}{27(r-2)^2} \leq \frac{25}{27} + \frac{5}{108} = \frac{35}{36}, \end{aligned} \quad (11)$$

where the last inequality uses $r-2 \geq 128$. The binomial neighbor ratio is

$$\rho_j = \frac{b_{j-1}b_{j+1}}{b_j^2} = \frac{j(N-j)}{(j+1)(N-j+1)}.$$

Since $a_j \geq 0$ and $\rho_j \geq 0$, we obtain

$$\begin{aligned} \frac{q_j^2 - q_{j-1}q_{j+1}}{b_j^2} &= (1 + a_j)^2 - \rho_j(1 + S_j) \\ &\geq 1 - \rho_j \left(1 + \frac{35}{36j}\right) \\ &= \frac{N + 35j + 36}{36(j+1)(N-j+1)} > 0. \end{aligned}$$

Every denominator is positive on the stated index range. This estimate is uniform in j , including all correction-support boundaries. $\square$

4 The finite interval and transfer of strictness

For every integer $5 \leq r \leq 129$, formula (5) computes the coefficients by integer arithmetic. Checking every internal difference requires exactly

$$\sum_{r=5}^{129} (3r - 6) = 24\,375$$

comparisons. All are strictly positive. Appendix A contains the full executable certificate, using arbitrary-precision integers and no external package. It covers all 125 integer parameters, without primality filtering, sampling, or symmetry reduction. For example, the minimum difference at $r = 5$ is 54, at $j = 9$ with adjacent coefficients (46, 10, 1); at $r = 129$ it is 73 153, at $j = 381$ with coefficients (72 771, 382, 1).

The finite interval was also recomputed in a separate mathematical review using polynomial convolution and binary exponentiation, without using the binomial-coefficient implementation. All 24 375 core inequalities agreed, and direct convolution checked all 32 875 internal inequalities of P_r in the same interval. These finite calculations establish only their stated interval; Proposition 5 supplies the entire range $r \geq 130$. Together they prove strict log-concavity of Q_r for all integers $r \geq 5$.

Lemma 6. *Let $a_0, \dots, a_n$ be positive, with $n \geq 2$, and strictly log-concave at every internal index. Extend the sequence by zero outside $[0, n]$. Then $d_j = a_j + a_{j-1}$, $0 \leq j \leq n+1$, is positive and strictly log-concave at every internal index.*

Proof. For $1 \leq j \leq n$, expansion gives

$$\begin{aligned} d_j^2 - d_{j-1}d_{j+1} &= (a_j^2 - a_{j-1}a_{j+1}) + (a_{j-1}^2 - a_{j-2}a_j) \\ &\quad + (a_ja_{j-1} - a_{j-2}a_{j+1}). \end{aligned}$$

The first two terms are nonnegative and at least one is positive, by internal strict log-concavity or the endpoint contributions a_0^2 and a_n^2 . The last term is nonnegative: when all indices are in the support, the nonincreasing successive ratios give

$$\frac{a_{j-1}}{a_{j-2}} \geq \frac{a_{j+1}}{a_j};$$

at the other indices its subtracted product is zero. $\square$

Completion of Theorems 1 and 2. The core Q_r has positive support of degree $3r - 5 \geq 10$. Apply Lemma 6 successively $r + 1$ times, one for each factor $1 + x$ in (3). This proves the asserted strict log-concavity of P_r at every $1 \leq j \leq 4r - 5$ for all integers $r \geq 5$. Equation (1) then proves the graph assertion for every prime $r \geq 5$. The quantitative core bound has already been proved in Proposition 5. $\square$

Computational and AI assistance. OpenAI Codex assisted with proof development, exact-computation code, mathematical review in a separate model context, and manuscript preparation. The finite calculations use exact integers; rational stress tests of the analytic bounds supplement, but do not replace, the proof above. The separate model review is neither external peer review nor proof-assistant formalization. The finite certificate below makes the computational step directly reproducible.

A Executable finite certificate

Save the following as `certificate.py` and run `python3 certificate.py` with Python 3.8 or later. Successful completion reports 125 parameters, 24375 strictly positive differences, and the minimum difference for each parameter. An explicit exception is raised on a failed inequality or a coverage mismatch; Python optimization does not disable these checks.

```

"""Finite core certificate. Run with Python 3.8+; standard library only."""
from math import comb
import json

def C(n, k):
    return comb(n, k) if 0 <= k <= n else 0

def core(r):
    return [C(3*r-5, k) + 2*C(2*r-4, k-1) + C(2*r-4, k-2)
            + 2*C(r-1, k-1) + C(r-1, k-2) + int(k == 1)
            for k in range(3*r-4)]

def verify():
    count = 0
    minima = []
    for r in range(5, 130):
        q = core(r)
        if len(q) != 3*r-4 or min(q) <= 0:
            raise ArithmeticError((r, "support"))
        delta = [q[j]**2 - q[j-1]*q[j+1]
                 for j in range(1, len(q)-1)]
        if min(delta) <= 0:
            raise ArithmeticError((r, "log-concavity"))
        count += len(delta)
        minima.append([r, min(delta)])
    if count != 24375 or len(minima) != 125:
        raise ArithmeticError("coverage")
    return {"parameters": 125, "inequalities": count,
            "all_positive": True, "minima": minima}

if __name__ == "__main__":
    print(json.dumps(verify(), indent=2))

```

References

- [1] B. A. Rather, *Combinatorial and analytic aspects of independence polynomials of zero divisor graphs*, arXiv:2606.04789v1 (2026). <https://arxiv.org/abs/2606.04789v1>.
- [2] Zijian Zeng, *The Distinct-Cycle-Length Probability is Strictly Decreasing after 30*, SSRN working paper (2026). <https://doi.org/10.2139/ssrn.7380519>.